**Web Data Integration**
**Final Report**

# Pharmaceutical Drug Integration: Linking US, EU and Canadian Approval Data

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# 1 Use Case

This project builds a unified view of approved pharmaceutical drugs across the United States, European Union, and Canada by integrating three official regulatory datasets: Drugs@FDA (US), EMA Medicines (EU), and the Health Canada Drug Product Database (DPD). The central entity is a medicine identified by its brand name and active substance(s); each integrated record includes licence holder, regulatory status, product class, ATC code, dosage form, route, strengths, substance count, regional dates, and generic/biosimilar/orphan flags.

# 2 Requirements Suitability

## 2.1 Schema and basic profile

Table 1: Overview of the input datasets

| Dataset | Source | Format | Rows | Entities | Attr. |
|---|---|---|---|---|---|
| Drugs@FDA | U.S. FDA | TXT/TSV | 51,839 | 8,175 | 56 |
| EMA Med. | EMA | XLSX/CSV | 2,746 | 2,648 | 39 |
| DPD | Health Can. | TXT/CSV | 42,042 | 41,636 | 87 |

Table 1 summarises the three input datasets. Drugs@FDA contains 51,839 product rows forming 8,175 entities across 29,377 applications (78.3% ANDAs). EMA Medicines contains 2,746 rows consolidated into 2,648 entities (including 28 records recovered via INN fallback), predominantly human medicines (85.6%). The Health Canada DPD contains 58,137 product records consolidated into 41,636 entities. High missingness in all three datasets is largely structural: optional fields, category-specific attributes, or regulatory-event dates that apply only to specific statuses.

## 2.2 Integrated schema and overlap with input schemata

The integrated schema contains 16 attributes (Table 2); eleven occur in at least two sources. `activeSubstances` and `strengths` are list-valued, while the remaining 14 attributes are single-valued (including the numeric attribute `activeSubstanceCount`). Strengths are represented as numeric value–unit pairs.

## 2.3 Entity Identification and Count Requirement

A medicine is identified by its `brandName` and `activeSubstances`. Contrary to initial assumptions, `licenceHolder` cannot be used as an identifying attribute because the same medicine is frequently marketed by different regional subsidiaries or unrelated licensing partners across jurisdictions (for example, *Verkazia* is held by `HARROW EYE` in the US, `Santen Oy` in the EU, and `APOTEX INC` in Canada).

To estimate how many medicines appear in multiple datasets based on exact matching, we build a normalized key (`medicine_id`) from `brandName` and `activeSubstances` by

Table 2: Attribute Intersection with Integrated Schema

| Attribute name | Attribute type | Datasets in which the attribute is found |
|---|---|---|
| `brandName` | string | Drugs@FDA, EMA Medicines, DPD |
| `activeSubstances` | list (string) | Drugs@FDA, EMA Medicines, DPD |
| `activeSubstanceCount` | numeric (integer) | Drugs@FDA (derived), EMA Medicines (derived), DPD |
| `licenceHolder` | string | Drugs@FDA, EMA Medicines, DPD |
| `regulatoryStatus` | string (categorical) | Drugs@FDA, EMA Medicines, DPD |
| `productClass` | string (categorical) | Drugs@FDA (derived), EMA Medicines, DPD |
| `atcCode` | string | EMA Medicines, DPD |
| `generic` | boolean | Drugs@FDA (derived), EMA Medicines |
| `biosimilar` | boolean | EMA Medicines |
| `orphanMedicine` | boolean | EMA Medicines |
| `dosageForm` | string | Drugs@FDA, DPD |
| `route` | string | Drugs@FDA, DPD |
| `strengths` | list (numeric value, unit) | Drugs@FDA, DPD |
| `usApprovalDate` | date | Drugs@FDA |
| `euAuthorisationDate` | date | EMA Medicines |
| `caStatusDate` | date | DPD |

converting text to upper case, removing accents and dosage strengths, standardizing punctuation, and sorting multiple active substances. Let $F$, $E$ and $D$ denote the sets of `medicine_id` values in Drugs@FDA, EMA Medicines and the DPD. The pairwise exact overlaps are $|F \cap E| = 428$, $|F \cap D| = 949$ and $|E \cap D| = 594$, yielding a deduplicated union of $|(F \cap E) \cup (F \cap D) \cup (E \cap D)| = 1{,}407$ distinct medicines – well above the required minimum of 1,000. (Note that this initial exploratory estimate was calculated on raw tabular extracts prior to final EMA entity consolidation; when evaluated via transitive graph clustering over the finalized XML datasets in Section 4.5, the exact baseline resolves to 1,372 unified multi-source medicine clusters).

However, exact matching misses many identical medicines because both identifying attributes can vary across countries (Table 3):

- **Active substance naming and salt forms:** The same substance can have different regional spellings (e.g., `CYCLOSPORINE` in the US and Canada vs. `ciclosporin` in the EU for *Verkazia*) or different chemical and salt representations (e.g., `GLYCOPYRROLATE` in the US, `glycopyrronium bromide` in the EU, and `GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE)` in Canada for *Seebri*).
- **Brand name suffixes:** The same medicine may carry different device or formulation suffixes across markets, such as `SEEBRI NEOHALER` in Drugs@FDA versus `Seebri Breezhaler` in EMA Medicines and the DPD.

## 3 Phase I: Data Selection and Translation

Phase I translates three pharmaceutical datasets into a shared XML target schema using Altova MapForce 2027. The outputs are validated XML files consumed by Phases II and III.

Table 3: Examples of identical medicines with differing identifying attributes across sources

| Dataset | Brand Name | Active Substance |
|---|---|---|
| Drugs@FDA | VERKAZIA | CYCLOSPORINE |
| EMA Medicines | Verkazia | ciclosporin |
| Health Canada DPD | VERKAZIA | CYCLOSPORINE |
| Drugs@FDA | SEEBRI NEOHALER | GLYCOPYRROLATE |
| EMA Medicines | Seebri Breezhaler | glycopyrronium bromide |
| Health Canada DPD | SEEBRI BREEZHALER | GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE) |

## 3.1 Target Schema

The target schema (`medicines.xsd`) represents one medicine entity per source record, identified by `brandName` and `activeSubstances`, with 16 attributes and two metadata fields (`id`, `source`). Three conventions apply across all mappers: missing values are omitted rather than written as empty tags; repeating values use typed wrapper elements (`activeSubstances`, `strengths`); and all dates follow ISO 8601 (YYYY-MM-DD). Controlled vocabularies are enforced for `regulatoryStatus` (six values: *Authorised, Suspended, Pending, Withdrawn, Application withdrawn, Refused*) and `productClass` (*Human, Veterinary, Disinfectant*).

## 3.2 Shared Helper Functions

A shared MapForce library (`wdi_helpers.mfd`) provides ten user-defined functions imported by all mappers (Table 4). Lookup XML files cover 25 regulatory statuses, 227 dosage forms, 147 routes, and 202 strength units across all three sources.

Table 4: Shared UDFs used across mappers

| Function | Purpose |
|---|---|
| `makeMedicineId` | Constructs a stable entity ID from source prefix and record key |
| `getActiveSubstances` | Tokenizes substance string; supports INN fallback for blank primary field |
| `standardizeRegulatoryStatus` | Maps raw status values to the six controlled vocabulary entries |
| `standardizeProductClass` | Maps category values to Human / Veterinary / Disinfectant |
| `standardizeAtcCodes` | Combines human and vet ATC codes; deduplicates, sorts, corrects typos |
| `yesNoToBoolean` | Converts Yes/No strings to `xs:boolean`; omits element when false |
| `parseAuthorisationDate` | Converts DD/MM/YYYY to ISO 8601 YYYY-MM-DD |
| `stdDosageForm` / `dosageFormList` | Standardizes single or multi-value dosage form strings |
| `stdRoute` / `routeList` | Standardizes single or multi-value administration routes |
| `splitStrengths` | Tokenizes a DPD strength string into (value, unit) pairs |

### 3.3 EMA Medicines Mapper

The EMA dataset originates from a flat table of 2,746 rows consolidated into 2,648 unique medicine entities (`ema_entities.csv`). Each consolidated entity row maps to one output `<medicine>` element. Because EMA does not publish strength or dosage form data in the register export, `dosageForm`, `route`, and `strengths` are structurally absent from EMA output. Table 5 lists all field correspondences.

Table 5: EMA Medicines field mapping

| Target field | Source column | Transformation |
|---|---|---|
| `id` | medicine_id | `makeMedicineId("EMA", id)` |
| `brandName` | Name of medicine | Direct copy |
| `activeSubstances` | Active substance / INN | `getActiveSubstances`; INN fallback |
| `activeSubstanceCount` | (derived) | `count(activeSubstances)` |
| `licenceHolder` | MA holder | Direct copy |
| `regulatoryStatus` | Medicine status | `standardizeRegulatoryStatus` |
| `productClass` | Category | `standardizeProductClass` |
| `atcCode` | ATC code (human) + ATCvet | `standardizeAtcCodes` |
| `generic` / `biosimilar` / `orphanMedicine` | Generic / Biosimilar / Orphan | `yesNoToBoolean` |
| `euAuthorisationDate` | Marketing authorisation date | `parseAuthorisationDate` |

The most intricate correspondence handles active substances. For 57 of the 2,746 rows, both `Active substance` and the `INN / common name` column are blank; because `activeSubstances` is mandatory in the target schema, these rows are excluded via a row filter. A further 28 rows have a blank primary field but a populated INN column: the mapper applies an INN fallback, normalizing whitespace and selecting the non-empty value. Commas and the word "and" are not treated as separators, preserving multi-word biological descriptions. Consolidating the 2,689 usable rows by normalized brand name and active substance set yields **2,648 unique medicine entities** (`ema_entities.csv`).

EMA maintains separate human and veterinary ATC code columns. `standardizeAtcCodes` concatenates both, strips asterisks used for provisional codes, drops the *Not yet assigned* placeholder, corrects a known EMA typo (`VO4D` → `V04D`), then deduplicates, sorts, and joins codes with `"; "`. Date conversion (`parseAuthorisationDate`) reorders DD/MM/YYYY to YYYY-MM-DD. The three boolean flags are stored as Yes/No strings; `yesNoToBoolean` converts them and omits the element when false, consistent with the `minOccurs="0"` schema convention. The full export passed XSD validation with **2,648 unique `EMA_` IDs** and zero duplicate or schema errors.

### 3.4 Drugs@FDA Mapper

The Drugs@FDA source consists of relational TSV tables in which multiple product rows across applications (`ApplNo`) correspond to the same medicine entity. Product records from `Products.txt` are joined to `Applications.txt` on `ApplNo`, linked to marketing status lookup values, and consolidated by normalized `brandName` (`DrugName`) and sorted `activeSubstances` (`ActiveIngredient`) into **8,175 unique medicine entities**. Eleven orphan product rows lacking a matching application and records with empty active ingredients (such as `ApplNo=221337`, `MEDICAL AIR, USP`) are excluded during

preparation. Table 6 summarises the field mappings.

Table 6: Drugs@FDA field mapping

| Target field | Source column | Transformation |
|---|---|---|
| `id` | medicine_id | `makeMedicineId("FDA", id)` |
| `brandName` | DrugName (plurality) | Direct copy |
| `activeSubstances` | ActiveIngredient | `getActiveSubstances`; separator `"; "` |
| `activeSubstanceCount` | (derived) | `count(activeSubstances)` |
| `licenceHolder` | SponsorName (plurality) | Direct copy |
| `regulatoryStatus` | MarketingStatusID | `standardizeRegulatoryStatus` |
| `productClass` | (constant) | `"Human"` |
| `generic` | ApplType (`ANDA`/`NDA`/`BLA`) | `true` if all `ANDA`; `false` if all `NDA`/`BLA`; omit if mixed |
| `dosageForm` | Form (form component) | `fdaFormPart` → `dosageFormList` |
| `route` | Form (route component) | `fdaRoutePart` → `routeList` |
| `strengths` | Strength | Pre-split combination tokens → `splitStrengths` |
| `usApprovalDate` | SubmissionStatusDate (`AP`) | Earliest non-empty approval date (ISO 8601) |

Drugs@FDA stores dosage form and administration route jointly in `Products.Form` (e.g. `"TABLET;ORAL"`); the helper functions `fdaFormPart` and `fdaRoutePart` split this composite field before lookup standardization. Combination strength strings such as `600MG/90MG` are pre-split into individual mass tokens prior to numeric value/unit extraction. Because Drugs@FDA covers only human drugs, `productClass` maps to *Human*. The `generic` flag is derived from application types: `true` when all contributing applications are `ANDA`, `false` when all are innovator applications (`NDA` or `BLA`), and omitted for the approximately 375 mixed entities. The resulting XML export (`fda_medicines.xml`) passed XSD and duplicate validation with **8,175 unique** `FDA_` **IDs**.

## 3.5 Health Canada DPD Mapper

The DPD source is a pre-consolidated CSV of **41,636 medicine entities** derived from 58,137 product records across 12 Health Canada tables (`dpd_entities.csv`). Unlike the EMA flat table, each DPD entity already aggregates multiple product records, so the mapper operates on one row per entity without further consolidation. The DPD source provides all 16 target attributes including `dosageForm`, `route`, and `strengths`, making it the most complete of the three sources.

The main mapping challenge in DPD is strength parsing. Each entity row carries a semicolon-delimited strength string (e.g. `"10MG; 0.5%"`); `splitStrengths` tokenizes this into individual tokens, from which `strengthValue` extracts the numeric component and `strengthUnit` normalizes the unit via `lookups/strength_unit.xml`. Tokens without a parseable numeric value are filtered out before the `<strength>` wrapper is created, and the wrapper itself is omitted when no valid tokens remain. Regulatory status presents a similar aggregation challenge: DPD entities can carry multiple historical statuses (e.g. `"MARKETED; CANCELLED POST MARKET"`), so `standardizeRegulatoryStatus` selects the highest active status according to the precedence hierarchy Authorised > Suspended > $\cdots$ > Refused.

Table 7: Health Canada DPD field mapping

| Target field | Source column | Transformation |
|---|---|---|
| `id` | medicine_id | `makeMedicineId("DPD", id)` |
| `brandName` | brand_name | Direct copy |
| `activeSubstances` | active_substances | `getActiveSubstances`; separator `"; "` |
| `activeSubstanceCount` | (derived) | `count(activeSubstances)` |
| `licenceHolder` | licence_holder | Direct copy |
| `regulatoryStatus` | current_status | `standardizeRegulatoryStatus` |
| `productClass` | product_class | `standardizeProductClass` |
| `atcCode` | atc_code | `standardizeAtcCodes` |
| `biosimilar` | biosimilar | `yesNoToBoolean` |
| `dosageForm` | dosage_forms | `dosageFormList` |
| `route` | routes | `routeList` |
| `strengths` | strengths | `splitStrengths` → (value, unit) pairs |
| `caStatusDate` | status_date | Direct copy (ISO date) |

The full export passed XSD validation with **41,636 records**, 100,094 strength pairs across 41,574 medicines, and no schema errors.

### 3.6 Integrated XML Validation Summary

All three translated XML files in `data/4_translated_mapforce/` were validated against `medicines.xsd` and checked for duplicate IDs, duplicate attribute content, and duplicate normalized entity keys using `check_schema.py` (Table 8). Together, the three sources provide **52,459 validated medicine entities** for Phase II identity resolution.

Table 8: Validation summary of translated XML datasets

| Validation check | `dpd_medicines.xml` | `ema_medicines.xml` | `fda_medicines.xml` |
|---|---|---|---|
| Records | 41,636 | 2,648 | 8,175 |
| Unique IDs | 41,636 | 2,648 | 8,175 |
| XSD Schema valid | True | True | True |
| Duplicate IDs | 0 | 0 | 0 |
| Duplicate content (excl. ID) | 0 | 0 | 0 |
| Duplicate normalized entities | 0 | 0 | 0 |

## 4 Phase II: Identity Resolution

Phase II resolves cross-jurisdictional medicine correspondences across Health Canada DPD (41,636 entities), EMA Medicines (2,648 entities), and Drugs@FDA (8,175 entities). Evaluating the full Cartesian product across the three dataset pairs would require $41{,}636 \times 2{,}648 + 41{,}636 \times 8{,}175 + 2{,}648 \times 8{,}175 = 472{,}273{,}828$ pairwise comparisons, motivating blocking, supervised and zero-shot matchers, and transitive connected-components clustering.

### 4.1 Gold-Standard Dataset Construction and Stratified Split

To construct a representative benchmark that tests both standard matches and adversarial pharmaceutical corner cases, candidate pairs for manual annotation were sampled

in two balanced pools using `gold_standard_pair_gen.py`:

- **Ordinary Pool (`hard=0`):** Contains candidate pairs with exact or high-similarity normalized brand and active substance keys alongside random negative pairs.
- **Hard Pool (`hard=1`):** Targets domain-specific corner cases where string similarity is misleading: (i) identical brand names with different active substances or combination vs. single-agent formulas, (ii) identical active substances marketed by competing generic manufacturers, and (iii) same medicine entities with regional INN salt variations or device/formulation suffixes.

Three team annotators manually reviewed all candidate batches across `DPD/EMA` (1,000 pairs), `DPD/FDA` (769 pairs), and `EMA/FDA` (1,729 pairs), yielding **3,498 unique reviewed pairs** with zero duplicate rows, zero conflicting labels, and zero unlabeled records. Two records were labeled as a match (`match` = 1) if and only if they share the same pharmacological active substance set (accounting for salt forms, hydration states, and regional INN spellings) and represent the same commercial medicine entity (including regional subsidiaries of the same corporate group, while excluding unrelated competing generic manufacturers).

Using `gold_standard_combine_split.py`, the 3,498 unified pairs were partitioned into an 85% development split (`gs_pair_train_val.csv`, **2,973 pairs**) and a 15% held-out test split (`gs_pair_test.csv`, **525 pairs**) stratified jointly by (`source_pair`, `hard`, `match`) with fixed seed 42 (Table 9).

Table 9: Stratified gold-standard dataset distribution across source pairs, difficulty pools, and partitions

| Source Pair | Difficulty Pool | Label | Whole | Train/Val (85%) | Test (15%) |
|---|---|---|---|---|---|
| DPD / EMA | Ordinary (`hard=0`) | Match (1) | 198 | 168 | 30 |
| DPD / EMA | Ordinary (`hard=0`) | Non-match (0) | 302 | 257 | 45 |
| DPD / EMA | Hard (`hard=1`) | Match (1) | 217 | 184 | 33 |
| DPD / EMA | Hard (`hard=1`) | Non-match (0) | 283 | 241 | 42 |
| DPD / FDA | Ordinary (`hard=0`) | Match (1) | 122 | 104 | 18 |
| DPD / FDA | Ordinary (`hard=0`) | Non-match (0) | 364 | 309 | 55 |
| DPD / FDA | Hard (`hard=1`) | Match (1) | 126 | 107 | 19 |
| DPD / FDA | Hard (`hard=1`) | Non-match (0) | 157 | 133 | 24 |
| EMA / FDA | Ordinary (`hard=0`) | Match (1) | 282 | 240 | 42 |
| EMA / FDA | Ordinary (`hard=0`) | Non-match (0) | 670 | 570 | 100 |
| EMA / FDA | Hard (`hard=1`) | Match (1) | 292 | 248 | 44 |
| EMA / FDA | Hard (`hard=1`) | Non-match (0) | 485 | 412 | 73 |
| **Total** | **All Pools** | **All Labels** | **3,498** | **2,973** | **525** |

## 4.2 Blocking and Candidate Generation

The unreduced search space across the three translated datasets encompasses 41,636 × 2,648 = 110,252,128 pairs for DPD–EMA, 41,636 × 8,175 = 340,374,300 pairs for DPD–FDA, and 2,648 × 8,175 = 21,647,400 pairs for EMA–FDA, yielding a total Cartesian product of **472,273,828 possible comparisons**. Performing detailed attribute matching or transformer inference across nearly half a billion pairs is computationally prohibitive. The goal of blocking is to filter out obvious non-matches while maintaining

high completeness on true correspondences. Performance is governed by two complementary metrics:

- **Reduction Ratio (RR):** $\text{RR} = 1 - \frac{|\mathcal{C}|}{|A \times B|}$, measuring the relative reduction in pairwise comparisons (target: $> 99.5\%$).
- **Pairs Completeness (PC):** $\text{PC} = \frac{|\mathcal{C} \cap \mathcal{M}|}{|\mathcal{M}|}$, the fraction of true matches $\mathcal{M}$ retained in candidate shortlist $\mathcal{C}$ (target: $> 97.0\%$ per source pair).

**Lexical Blocking Baselines.** We initially benchmarked standard lexical blocking rules implemented via PyDI (`StandardBlocker` and `TokenBlocker`) on the development split (`gs_pair_train_val.csv`, 1,038 unique eligible positive matches after EMA ID alignment and deduplication; Table 10):

1. **Brand Prefix-4:** Compares the first four alphanumeric characters of normalized brand names. While achieving high reduction ratios ($> 99.96\%$), it missed 39 of the 352 DPD–EMA matches (PC = 88.92%) due to regional brand prefixes (e.g., `TEVA-`, `APO-`, or formulation prefixes).
2. **Brand Trigrams (capped at 1,000):** Tokenizes brand names into overlapping 3-character shingles. To prevent Cartesian explosions on frequent pharmaceutical fragments (e.g., `"TAB"`, `"SOL"`), candidate generation was capped at $a \times b \leq 1{,}000$ co-occurrences per shingle. However, trigrams alone missed 39 matches on DPD–FDA (PC = 81.52%).
3. **Prefix-4 $\cup$ Trigrams:** Combining both lexical rules improved completeness to $\geq 98.30\%$ on all pairs, but generated 1,369,537 candidates (663,705 on DPD–FDA alone), representing a substantial downstream processing burden.

**Dense Semantic Embeddings (Qwen3-Embedding-0.6B).** To capture semantic equivalence across international nonproprietary names (INNs), salt variations, and brand synonyms, we evaluated dense vector blocking using `Qwen/Qwen3-Embedding-0.6B` (1,024-dimensional representations). For each medicine entity, the normalized brand name and alphabetically sorted active substances were concatenated and encoded without fine-tuning. Cosine nearest-neighbor search was executed from the smaller source into the larger source. A shortlist of the top-5 nearest neighbors per medicine produced only **67,355 total candidates** while retaining 1,021 of 1,038 development positives (**98.36% pooled completeness**; Table 10).

**Frozen Evaluation and Final Matcher Handoff.** When Qwen Top-5 was evaluated on the held-out test set (185 known matches), it achieved 98.41% on DPD–EMA and 97.30% on DPD–FDA, but dropped to 92.94% (79/85) on EMA–FDA (95.68% pooled completeness), failing the per-source 97% target. Diagnostic error inspection revealed that missed EMA–FDA pairs shared brand prefixes that lexical prefix matching reliably captures.

Table 10: Comparison of lexical and dense embedding blocking methods on the development set (1,038 positive matches)

| Source Pair | Blocking Method | Candidates | Matches | Retained | Completeness | Red. Ratio |
|---|---|---|---|---|---|---|
| DPD–EMA | Brand Prefix-4 | 16,398 | 352 | 313 | 88.92% | 99.9851% |
| DPD–EMA | Brand Trigrams (cap 1,000) | 431,353 | 352 | 323 | 91.76% | 99.6088% |
| DPD–EMA | Prefix-4 ∪ Trigrams | 443,207 | 352 | 346 | 98.30% | 99.5980% |
| DPD–EMA | Qwen Top-5 | 13,240 | 352 | 346 | 98.30% | 99.9880% |
| DPD–EMA | **Qwen Top-5 ∪ Prefix-4** | **27,966** | **352** | **350** | **99.43%** | **99.9746%** |
| DPD–FDA | Brand Prefix-4 | 133,069 | 211 | 210 | 99.53% | 99.9609% |
| DPD–FDA | Brand Trigrams (cap 1,000) | 544,036 | 211 | 172 | 81.52% | 99.8402% |
| DPD–FDA | Prefix-4 ∪ Trigrams | 663,705 | 211 | 210 | 99.53% | 99.8050% |
| DPD–FDA | Qwen Top-5 | 40,875 | 211 | 210 | 99.53% | 99.9880% |
| DPD–FDA | **Qwen Top-5 ∪ Prefix-4** | **162,930** | **211** | **211** | **100.00%** | **99.9521%** |
| EMA–FDA | Brand Prefix-4 | 4,456 | 475 | 468 | 98.53% | 99.9794% |
| EMA–FDA | Brand Trigrams (cap 1,000) | 261,601 | 475 | 472 | 99.37% | 98.7915% |
| EMA–FDA | Prefix-4 ∪ Trigrams | 262,625 | 475 | 474 | 99.79% | 98.7868% |
| EMA–FDA | Qwen Top-5 | 13,240 | 475 | 465 | 97.89% | 99.9388% |
| EMA–FDA | **Qwen Top-5 ∪ Prefix-4** | **15,934** | **475** | **474** | **99.79%** | **99.9264%** |

To guarantee compliance with the project's $\geq 97\%$ completeness requirement across all three pairs, the final handoff rule was defined as the union:

$$\mathcal{C}_{\text{final}} = \mathcal{C}_{\text{Qwen top-5}} \cup \mathcal{C}_{\text{Prefix-4}}$$

As shown in Table 11, this hybrid rule recovers 6 additional test matches, bringing EMA–FDA to **97.65% completeness** and achieving **100.00% completeness** on DPD–EMA and DPD–FDA (**98.92% pooled completeness**, 183/185). The candidate shortlist totals **206,830 unique pairs**, compressing the search space by **99.9562%** (over 2,280× reduction). All candidates were persisted to `data/7_blocking_candidates/` with cryptographic SHA-256 checksums verified in `manifest.json`.

Table 11: Final candidate handoff summary and held-out test set evaluation (185 known matches)

| Source Pair | Cartesian Space | Candidates | Red. Ratio | Test Matches | Retained | Completeness |
|---|---|---|---|---|---|---|
| DPD–EMA | 110,252,128 | 27,966 | 99.9746% | 63 | 63 | **100.00%** |
| DPD–FDA | 340,374,300 | 162,930 | 99.9521% | 37 | 37 | **100.00%** |
| EMA–FDA | 21,647,400 | 15,934 | 99.9264% | 85 | 83 | **97.65%** |
| **Pooled** | **472,273,828** | **206,830** | **99.9562%** | **185** | **183** | **98.92%** |

## 4.3 Rule-Based, Machine Learning, and Fine-Tuned PLM Matchers

## 4.4 Zero-Shot LLM Matcher Benchmark

To evaluate whether a pretrained Large Language Model can resolve cross-jurisdictional pharmaceutical entities without task-specific training examples, we implemented a strictly blind Zero-Shot LLM matcher (`src/identity_resolution/llm_zero_shot_matcher.py`) using Google's `gemini-3.8-flash` model via the official `google-genai` SDK (`temperature` = 0.0, structured JSON schema output enforcing integer `match` $\in \{0, 1\}$ and clinical rationale string `reason`).

**Decontaminated Zero-Shot Prompt Architecture.** Each test pair in `gs_pair_test.csv` ($n = 525$) is formatted with four record attributes per jurisdiction: `brandName`, `activeSubstances`, `licenceHolder`, and `dosageForm` (with missing values formatted as `"Not specified"`). The system instruction encodes three general domain rules without exposing any test-set drug names or few-shot examples:

1. **Active Substances:** Ingredients must represent the same pharmacological moiety; salt forms (e.g. hydrochloride, sodium, maleate, besylate), hydration states, and INN vs. USAN spellings (e.g. ciclosporin vs. cyclosporine) are treated as equivalent, whereas single-substance vs. multi-substance combination products are non-matches.

2. **Brand Name and Licence Holder:** Identical brands or minor formulation suffixes from regional subsidiaries of the same corporate group are matches, whereas generic products from unrelated competing manufacturers are non-matches.

3. **Dosage Form Variations:** Minor dosage form differences do not invalidate a match when brand name, active substance set, and corporate lineage agree.

**Comparison Against the Naive Exact-Match Baseline.** We benchmark the Zero-Shot LLM matcher against a deterministic **Naive Exact-Match Baseline** that predicts `match` = 1 if and only if normalized `brandName1` equals `brandName2` and their normalized sets of semicolon-delimited `activeSubstances` are identical. Table 12 reports overall test-set performance across all 525 held-out pairs.

Table 12: Held-out test set evaluation ($n = 525$ pairs): Naive Exact-Match Baseline vs. Zero-Shot LLM (`gemini-3.8-flash`)

| Metric | Exact-Match Baseline | Zero-Shot LLM (`gemini-3.8-flash`) | Improvement (Δ) |
|---|---|---|---|
| **Macro F1 Score** | 0.7307 (73.07%) | **0.9485 (94.85%)** | **+21.78 pp** |
| **Match Class F1 (1)** | 0.6122 (61.22%) | **0.9344 (93.44%)** | **+32.21 pp** |
| **Non-Match Class F1 (0)** | 0.8492 (84.92%) | **0.9626 (96.26%)** | **+11.34 pp** |
| **Overall Accuracy** | 0.7829 (78.29%) | **0.9524 (95.24%)** | **+16.95 pp** |
| Macro Precision | 0.8016 (80.16%) | 0.9443 (94.43%) | +14.27 pp |
| Macro Recall | 0.7154 (71.54%) | 0.9534 (95.34%) | +23.80 pp |
| Match Precision (Class 1) | 0.8333 (83.33%) | 0.9128 (91.28%) | +7.95 pp |
| Match Recall (Class 1) | 0.4839 (48.39%) | 0.9570 (95.70%) | +47.31 pp |
| True Positives (TP) | 90 | **178** | +88 matches found |
| False Negatives (FN) | 96 | **8** | −88 missed matches |
| False Positives (FP) | 18 | **17** | −1 false alarm |
| True Negatives (TN) | 321 | **322** | +1 |

The Naive Exact-Match Baseline achieves a Macro F1 of 73.07% and misses more than half of all true matches ($\text{Recall}_1 = 48.39\%$, FN = 96) because regional salt modifiers and brand suffixes break exact string equality. By contrast, `gemini-3.8-flash` achieves **94.85% Macro F1** (95.24% accuracy), recovering 88 additional true matches and reducing false negatives by 91.7% (from 96 to 8).

**Performance by Difficulty Pool and Source Jurisdiction Pair.** Table 13 disaggregates test-set performance across difficulty pools and regulatory pair combinations. On the adversarial **Hard Pool** (`hard` = 1, 235 pairs), the Exact-Match Baseline fails completely on positive matches (Match F1 = 0.0000, Macro F1 = 0.3717), whereas the Zero-Shot LLM maintains **0.9296 Macro F1** (0.9167 Match F1, 93.19% accuracy). Across source pairs, the LLM achieves 0.9864 Macro F1 on `DPD/EMA`, 0.9704 on `DPD/FDA`, and 0.9144 on `EMA/FDA`.

Table 13: Zero-Shot LLM vs. Naive Baseline breakdown by difficulty pool and source jurisdiction pair

| Subgroup | Pairs | Base. Macro F1 | Base. Match F1 | LLM Acc. | LLM Macro F1 | LLM Match F1 |
|---|---|---|---|---|---|---|
| *Breakdown by Difficulty Pool* | | | | | | |
| Ordinary Pool (`hard=0`) | 290 | 0.9310 | 0.9091 | 96.90% | **0.9647** | 0.9524 |
| Hard Pool (`hard=1`) | 235 | 0.3717 | 0.0000 | 93.19% | **0.9296** | 0.9167 |
| *Breakdown by Source Jurisdiction Pair* | | | | | | |
| DPD / EMA (Canada ↔ EU) | 150 | 0.7366 | 0.6383 | 98.67% | **0.9864** | 0.9844 |
| DPD / FDA (Canada ↔ US) | 116 | 0.6979 | 0.5625 | 97.41% | **0.9704** | 0.9600 |
| EMA / FDA (EU ↔ US) | 259 | 0.7408 | 0.6176 | 92.28% | **0.9144** | 0.8876 |

**Qualitative Clinical Error Analysis.** Inspecting all 25 test-set discrepancies (17 False Positives and 8 False Negatives) reveals four domain patterns:

1. **Cross-Border Licensing and Corporate Acquisitions (14 FP):** In 14 pairs with identical brand names and active substances (e.g. *Xolair* marketed by Novartis in the EU and Genentech in the US; *Doptelet* by SOBI vs. wholly owned subsidiary AkaRx; *Tekturna* by Novartis vs. acquirer LXO; *Nevanac* by Novartis vs. Harrow Eye; *Adasuve* by Ferrer vs. Nova Pneuma; *Zurampic* by Grünenthal vs. Ironwood), the LLM leveraged pretrained domain knowledge of licensing partnerships to predict `match` = 1, whereas annotators labeled them 0 due to differing surface manufacturer names (plus *Tevimbra* by BeOne Medicines Germany vs. Ireland, which is a gold-standard annotation error).

2. **Formulation Suffix Variants (3 FP, 1 FN):** Modified delivery formulations from the same innovator (*Natpar* vs. *Natpara*, *Alphagan* vs. *Alphagan P*, and *Afinitor Disperz* vs. *Afinitor* predicted 1 vs. GS 0; *Tobi Podhaler* dry powder inhaler vs. *TOBI* nebulizer solution predicted 0 vs. GS 1).

3. **Regional Co-Brands, Prodrugs, and Co-Formulations (3 FN):** *Copalia HCT* vs. *Exforge HCT* (distinct EU co-marketing brand names from Novartis), *Ivemend* (`fosaprepitant` intravenous prodrug) vs. *Emend* (`aprepitant` oral parent drug), and *Herceptin* (`trastuzumab`) vs. *Herceptin Hylecta* (`trastuzumab and hyaluronidase`).

4. **Gold-Standard Label Noise Corrected by the LLM (4 FN):** Four pairs labeled 1 in the manual gold standard are genuine non-matches where the LLM's 0 prediction is clinically correct: `AURO-LAMIVUDINE/ZIDOVUDINE` (dual therapy)

vs. `EFAVIRENZ; LAMIVUDINE; ZIDOVUDINE` (triple therapy), and three competing generic pairs (*Repaglinide Accord* vs. Aurobindo, *Pemetrexed Krka* vs. Shilpa, and *Tadalafil Lilly* vs. Rising).

### 4.5 Tri-Dataset Connected-Components Clustering

Pairwise entity resolution produces binary correspondence sets across `DPD` ↔ `EMA`, `DPD` ↔ `FDA`, and `EMA` ↔ `FDA`. To consolidate these pairwise links into unified multi-source medicine entities for Phase III fusion, we implemented graph-based transitive closure in `src/identity_resolution/clustering.py` using NetworkX connected components. Let $G = (V, E)$ be an undirected graph where vertices $V$ are source-prefixed entity IDs (`DPD_*`, `EMA_*`, `FDA_*`) and edges $E$ are positive pairwise match correspondences. Each connected component $C_k \subseteq V$ forms a unified medicine cluster assigned a deterministic identifier (`CLUSTER_00001`, ...).

Executing `clustering.py` on the exact normalized baseline correspondences across the three validated XML datasets produces **1,372 cross-dataset medicine clusters** spanning at least two jurisdictions, including **283 three-way clusters** present in Canada, the EU, and the US simultaneously (Table 14). This confirms that the integrated dataset exceeds the course threshold of 1,000 overlapping entities even before incorporating fuzzy and learned matcher correspondences. This count refines the preliminary exploratory profiling estimate of 1,407 reported in Section 2.3: whereas the exploratory estimate counted distinct synthetic keys across raw pre-consolidation CSV extracts, connected-components clustering operates directly on the deduplicated XML target datasets (where EMA was consolidated into 2,648 unique entities) and applies transitive graph closure to unify pairwise correspondences into coherent multi-source entity clusters.

Furthermore, when clustering is executed over the full candidate pool generated by the hybrid blocker ($\mathcal{C}_{\text{Qwen top-5}} \cup \mathcal{C}_{\text{Prefix-4}}$, 206,830 candidate pairs) using our hierarchical matcher with salt-form normalization and deterministic greedy 1:1 conflict resolution (`--enforce-one-to-one`), the yield of cross-dataset medicines increases to **2,343 clusters** (+70.8% gain over the exact baseline), with three-way international clusters increasing to **502** (+77.4% gain).

**Cluster / Group Size Distribution.** A crucial quality metric for identity resolution is the group size distribution resulting from transitive closure over match correspondences. Table 14 summarizes the resulting cluster sizes for both the Exact Baseline and the Candidate Correspondences.

- **Exact Baseline:** Transitive closure unifies 3,027 medicine records into 1,372 clusters (mean size 2.21). Exactly 1,089 clusters (79.37%) are pairwise ($k = 2$, linking two jurisdictions), and 283 clusters (20.63%) are triplets ($k = 3$, linking Canada, the EU, and the US). No clusters exceed size 3 (100.0% pure 1:1 and 1:1:1 equivalence classes with zero intra-source conflicts).

Table 14: Connected-components clustering profile: Group size distribution and jurisdictional overlap breakdown (Baseline Exact vs. Candidate Matches with Greedy 1:1)

| Metric / Category | Exact Baseline | Share | Candidate Matches | Share |
|---|---|---|---|---|
| ***Group Size Distribution (k)*** | | | | |
| Pairwise ($k = 2$) | 1,089 | 79.37% | 1,841 | 78.57% |
| Tri-Jurisdictional ($k = 3$) | 283 | 20.63% | 497 | 21.21% |
| Size $k = 4$ | 0 | 0.00% | 4 | 0.17% |
| Size $k = 5$ | 0 | 0.00% | 1 | 0.04% |
| **Total Multi-Source Clusters** | **1,372** | **100.00%** | **2,343** | **100.00%** |
| Total Entities Clustered | 3,027 | — | 5,194 | — |
| Intra-Source Variations ($k > 3$) | 0 | 0.00% | 5 | 0.21% |
| ***Jurisdictional Overlap Breakdown ($\geq 2$ Sources)*** | | | | |
| All 3 Sources (`DPD + EMA + FDA`) | 283 | 20.63% | 502 | 21.43% |
| Canada & US Only (`DPD + FDA`) | 629 | 45.85% | 1,388 | 59.24% |
| Canada & EU Only (`DPD + EMA`) | 311 | 22.67% | 314 | 13.40% |
| EU & US Only (`EMA + FDA`) | 149 | 10.86% | 139 | 5.93% |
| **Total Overlapping Entities** | **1,372** | **100.00%** | **2,343** | **100.00%** |

- **Candidate Matches (Greedy 1:1):** Transitive closure consolidates 5,194 entity records into 2,343 multi-source clusters (mean size 2.22). Pairwise clusters account for 1,841 clusters (78.57%), and tri-jurisdictional triplets account for 497 clusters (21.21%). Only 5 clusters (0.21%) contain more than three records (four of size 4, one of size 5), demonstrating that 99.79% of resolved candidate clusters are strictly duplicate-free single-entry-per-source components.

**Analysis of Clusters with Size $>$ 3.** Following course guidelines, clusters containing multiple entities from the same source jurisdiction were manually audited to assess whether they originate from false-positive transitive chaining or legitimate intra-source duplicates:

1. `CLUSTER_00001` ($k = 5$): Fixed-dose combination *Irbesartan and Hydrochlorothiazide*. Both EMA and FDA contain separate marketing authorizations for competing generic manufacturers (EMA has BMS and Teva licences; FDA has two distinct ANDA applications). Transitive closure links these duplicate generic authorisations to the single Canadian DPD entry, yielding 5 records.

2. `CLUSTER_00002` ($k = 4$): *Temozolomide*, an oncology therapeutic. Health Canada DPD contains two separate active DIN approvals from competing generic sponsors (*ACH Temozolomide* and *Temozolomide Capsules*) linking to single EMA and FDA records.

3. `CLUSTER_00003` ($k = 4$): *Sitagliptin + Metformin Hydrochloride*. DPD contains two distinct generic DIN approvals (*Apo-Sitagliptin* and *Reddy-Sitagliptin*) matching the corresponding EMA and FDA entries.

4. `CLUSTER_00004` ($k = 4$): *Emtricitabine + Tenofovir Disoproxil*. DPD registers two distinct generic approvals (*Jamp* and *Mylan*) clustering with the international reference entries.

5. `CLUSTER_00005` ($k = 4$): *Koselugo* (*selumetinib*). US Drugs@FDA lists two separate application records reflecting base substance (*selumetinib*) versus sulfate salt (*selumetinib sulfate*) representations, both clustering with the Canadian and European entries.

In all five cases, the expanded group size reflects legitimate upstream intra-source generic duplicates or multi-application salt variations in the regulatory registers rather than false-positive cross-substance matching errors.

# 5 Phase III: Data Fusion

Phase III integrates attributes across clustered entities, resolves conflicting values, and measures the quality profile of the consolidated pharmaceutical dataset. In accordance with the course report guidelines, this section documents the experimental results across six dimensions:

## 5.1 Provenance Tracking

Each fused entity and individual attribute value is enriched with provenance metadata, recording the source regulatory registry (`Drugs@FDA`, `EMA`, `DPD`), original national identifiers, and dataset release dates (Drugs@FDA 2026, EMA 2026, DPD 2026) to provide full lineage and auditability.

## 5.2 Dataset Density and Consistency Assessment

We evaluate the density of each input dataset prior to fusion alongside the merged dataset across the 16 target attributes. Intra-source and cross-source attribute consistency is analyzed to identify systematic discrepancies across jurisdictions (e.g., brand naming suffixes, salt representations, and regulatory status categories).

## 5.3 Fusion Evaluation Benchmark Set

To rigorously assess conflict resolution quality, an evaluation benchmark set containing $\geq 15$ multi-source medicine entities with $\geq 4$ attributes per entity is established. Ground-truth values are independently validated against external regulatory monographs and clinical pharmacopeia (e.g., DailyMed, WHO INN register), ensuring unambiguous single-record references without evaluation ambiguities.

## 5.4 Conflict Resolution Functions and Custom Rules

We systematically experiment with standard conflict resolution strategies (Voting, Longest String, Most Trusted Source based on jurisdiction data quality scores, and Union for multi-valued active substances and strengths) as well as custom pharmaceutical domain heuristics (e.g., INN-preference for substance nomenclature and hierarchical dosage form mapping).

### 5.5 Conflict Resolution Accuracy Comparison

The performance of different resolution functions is comparatively evaluated against the benchmark set. A comparative table reports attribute-level accuracy across alternative functions, identifying the optimal strategy for each target attribute.

### 5.6 Remaining Error Analysis

An error audit is performed on the remaining discrepancies following the application of the best resolution functions, categorizing residual errors into regulatory classification differences, temporal update lags, and typographical reporting variants.

## 6 Conclusion

In accordance with the project report requirements, the overall integration results are summarized by addressing three core evaluation questions:

1. **Additional Entities Added:** The total number of unique medicine entities gained beyond the largest individual input dataset (Health Canada DPD with 41,636 entities), quantifying the expanded international coverage achieved through tri-dataset alignment.

2. **Data Density Increase:** The net increase in attribute density achieved in the fused dataset relative to the highest-density input source, demonstrating the complementary informational yield of multi-jurisdiction merging.

3. **Overall Dataset Accuracy:** The end-to-end accuracy of the final fused dataset as measured against the independent ground-truth evaluation benchmark.

## Sources


- U.S. Food and Drug Administration, *Drugs@FDA Data Files*. https://www.fda.gov/drugs/drug-approvals-and-databases/drugsfda-data-files

- European Medicines Agency, *Download medicine data*. https://www.ema.europa.eu/en/medicines/download-medicine-data

- Health Canada, *Drug Product Database*. https://www.canada.ca/en/health-canada/services/drugs-health-products/drug-products/drug-product-database.html